

O'tgan yillar savollari

9-, 10- va 11-sinf

Задачи прошлых лет · 9, 10 и 11 классы

Qo'limizdagi barcha variantlar bitta faylda — asl sahifalari bilan, qayta terilmagan. Savollar matni ham, chizmalari ham original.

Все имеющиеся варианты в одном файле — оригинальные страницы, без перенабора. И текст задач, и чертежи — подлинные.

8 ta variant · 20 sahifa savol

2024-yil · 2025/26 tuman bosqichi (A va B variantlari)

Har bir variant: 30 savol · 90 daqiqa · 50 ball

Variantlar

9-sinf	2024	1–10: 0,9 ball · 11–20: 1,5 · 21–30: 2,6 — 21–30 ochiq javobli 1–10: 0,9 балла · 11–20: 1,5 · 21–30: 2,6 — 21–30 с открытым ответом	3
9-sinf	2025/26-A	1–20 yopiq (A–D) · 21–30 ochiq javobli 1–20 закрытые (A–D) · 21–30 с открытым ответом	5
9-sinf	2025/26-B	1–30 yopiq (A–D) 1–30 закрытые (A–D) Qoʻlda 3 sahifa bor, 4-sahifasi yoʻq — 30-savol tushib qolgan. В наличии 3 страницы, 4-й нет — задача 30 отсутствует.	8
10-sinf	2024	1–10: 0,9 ball · 11–20: 1,5 · 21–30: 2,6 — 21–30 ochiq javobli 1–10: 0,9 балла · 11–20: 1,5 · 21–30: 2,6 — 21–30 с открытым ответом	10
10-sinf	2025/26-A	1–20 yopiq (A–D) · 21–30 ochiq javobli 1–20 закрытые (A–D) · 21–30 с открытым ответом	12
10-sinf	2025/26-B	1–30 yopiq (A–D) 1–30 закрытые (A–D)	15
11-sinf	2024	1–10: 0,9 ball · 11–20: 1,5 · 21–30: 2,6 — 21–30 ochiq javobli 1–10: 0,9 балла · 11–20: 1,5 · 21–30: 2,6 — 21–30 с открытым ответом	18
11-sinf	2025/26-A	1–20 yopiq (A–D) · 21–30 ochiq javobli 1–20 закрытые (A–D) · 21–30 с открытым ответом	20

Eslatma. 2025/26-A va 2025/26-B — oʻsha tuman bosqichining ikki xil varianti: A da 20 ta yopiq va 10 ta ochiq savol, B da esa 30 tasi ham yopiq. Ikkalasining muqovasida ham 2025–2026-oʻquv yili yozilgan. Javoblar va toʻliq yechimlar alohida fayllarda: **Matematika-9/10/11-yechimlar-UZ-RU.pdf**.

1-qism: Har bir topshiriq 0,9 balldan baholanadi

- Ifodani soddalashtiring: $\frac{9a^2-24a+16}{9a^2-16} \cdot \frac{9a^2+24a+16}{9a^2+16}$
 A) $\frac{9a^2-24a+16}{9a^2+16}$ B) $\frac{9a^2-16}{9a^2+16}$ C) $\frac{3a+4}{3a-4}$ D) 1
- Tenglamani eng katta ildizini toping: $2x^2 - x(\sqrt{5} + 2) = -\sqrt{5}$
 A) 1 B) $2\sqrt{5}$ C) $\sqrt{5}$ D) $0,5\sqrt{5}$
- 80 ta konfet shuncha pul turadiki, 20 so'mga shuncha dona konfet sotib olish mumkin. 100 so'mga necha dona konfet sotib olish mumkin?
 A) 50 B) 150 C) 200 D) 120
- Ifodani soddalashtiring: $\frac{c-8}{\sqrt[3]{c^2+2\sqrt[3]{c}+4}} - \frac{c+8}{\sqrt[3]{c^2-2\sqrt[3]{c}+4}}$ A) $2\sqrt[3]{c}$ B) 4 C) -4 D) $2\sqrt[3]{c} - 4$
- Tenglamani haqiqiy ildizlari sonini toping. $(3x - 1)(3|x| - 1) = 3$
 A) 1 B) 2 C) 3 D) $\emptyset$
- $\begin{cases} \frac{a-b}{b} + 1 = 2 \\ \frac{a^2-b^2}{b^2} + 2 = c \end{cases}$ bo'lsa c ni toping. A) 5 B) 4 C) 3 D) 2
- $x - 1 \geq \frac{2024}{x+1}$ tengsizlikni qanoatlantiradigan eng kichik butun x sonini toping.
 A) -44 B) -45 C) 46 D) 45
- Tenglamani yeching $(12^2 + 6^2)(14^2 + 7^2) = 35^2 \cdot x^2$
 A) ± 10 B) ± 5 C) ± 6 D) ± 7
- Uchta ketma-ket kelgan toq natural sonlarning yig'indisi quyidagilardan qaysi biriga teng bo'la olmaydi? A) 12021 B) 20103 C) 30303 D) 40005
- Arifmetik progressiyaning dastlabki to'rtta hadi yig'indisi 30 ga teng va $a_9 - a_5 = 12$. Agar $a_n = 93$ bo'lsa, n nimaga teng? A) 29 B) 30 C) 31 D) 32

2-qism: Har bir topshiriq 1,5 balldan baholanadi

- $\overline{xy0xy0}$ soni x va y ning qandaydir qiymatlarida $(\overline{xyxy} + 9\overline{xy})$ ga bo'linadi. Bo'linma nechaga teng? A) 91 B) 90 C) 110 D) 101
- $2024^{20} \cdot 2025^{24} \cdot 20^{24}$ ko'paytma nechta nol bilan tugaydi?
 A) 48 B) 72 C) 68 D) 44
- $f(1 - 2x) = 1 - 2x^2$ bo'lsa, $f\left(-\frac{1}{2}\right) + f(0) + f\left(\frac{1}{2}\right)$ yig'indini hisoblang.
 A) 0 B) $1\frac{1}{8}$ C) $1\frac{1}{4}$ D) 3
- $x^2 + ax + 2024b = 0$ va $x^2 + bx + 2024a = 0$ tenglamalarning bitta ildizi umumiy. Agar $a \neq b$ bo'lsa, umumiy ildiz nechaga teng bo'lishi mumkin?
 A) $\frac{1}{2024}$ B) $-\frac{1}{2024}$ C) -2024 D) 2024
- O'tkir burchakli ABC uchburchakning balandliklari H nuqtada kesishadi. Agar $\sqrt{3}AB = CH$ ekani ma'lum bo'lsa, ACB burchakni toping.
 A) 45° B) 75° C) 30° D) 60°



16. $A(-1; -30)$ nuqtadan $y = 3(x^2 - 6x)$ parabola uchigacha bo'lgan masofani toping.
Найдите расстояние от точки $A(-1; -30)$ до вершины параболы $y = 3(x^2 - 6x)$.
A) $\sqrt{22}$ B) $\sqrt{30}$ C) 4 D) 5
17. Uchburchakning ikkita tomoni 15 va 13 ga teng, uchinchi tomoniga tushirilgan mediana esa 7 ga teng bo'lsa, berilgan uchburchak yuzini toping. A) 80 B) 72 C) 65 D) 84
18. $\sqrt{-x} + \sqrt{12 - x - x^2} \geq 3$ tengsizlikning butun sonlardan iborat yechimlar sonini toping. A) 5 B) 3 C) 2 D) 4
19. Gipotenuzasi 8 ga, gipotenuzaga tushirilgan balandligi $1\frac{1}{16}$ ga teng bo'lgan to'g'ri burchakli uchburchak perimetrini toping. A) 18 B) $18\frac{1}{8}$ C) 17 D) $17\frac{7}{8}$
20. Agar Алишер 3 kun, Anvar esa 5 kun ishlasa ishning 36%i bajariladi. Agar Alisher 5 kun, Anvar esa 3 kun ishlasa ishning 46%i bajariladi. Alisher va Anvar birgalikda 4 kun ishlasa ishning qanday qismi bajariladi? A) 39% B) 38% C) 40% D) 41%

3-qism: Har bir topshiriq 2,6 balldan baholanadi

21. x va y sonlari $x(1 + y) = y(1 - x) = 1$ tenglikni qanoatlantiradi.
 $(x - x^2)(y + y^2)(x + y)$ ifodaning qiymatini toping
22. $[2u] - \{u\} = 20,24$ tenglamaning ildizlari yig'indisini toping, bunda $[u]$ – u sonining butun qismi, $\{u\}$ – u sonining kasr qismi.
23. Perimetri 36 ga va yuzi 54 ga teng bo'lgan to'g'ri to'rtburchakning diagonallari orasidagi o'tkir burchakni toping.
24. a, b, c – turli raqamlar, $\overline{abbc}$ – to'rtxonal son va $\overline{abbc} = (9c + 2)^2$ tenglik o'rinli bo'lsa, $a + b + c$ ni toping.
25. $z + \sqrt{z + \frac{1}{2}} + \sqrt{z + \frac{1}{4}} = 2025$ tenglamani yeching.
26. ABC uchburchakning AQ va BP bissektrisalari O nuqtada kesishadi ($P \in AC, Q \in BC$).
Agar $\angle ACB = 50^\circ$ va $AP + OP = AB$ bo'lsa, $\angle ABC$ ni toping.
27. a, b, c, d – haqiqiy sonlar $\frac{a}{b+c+d} + \frac{b}{c+d+a} + \frac{c}{d+a+b} + \frac{d}{a+b+c} = 1$ tenglikni qanoatlantirsa, $\frac{a^2}{b+c+d} + \frac{b^2}{c+d+a} + \frac{c^2}{d+a+b} + \frac{d^2}{a+b+c}$ ni toping.
28. $ABCD$ qavariq to'rtburchakda $\angle ABC = \angle ADC = 90^\circ$. E, F nuqtalar mos ravishda AC va BD diagonallarning o'rtalari. $\angle BFE$ burchakni toping.
29. $\frac{x^2 - 2px + p^2 + p - 12}{x} = 0$ tenglama yagona yechimga ega bo'ladigan p parametrning qiymatlari yig'indisini toping.
30. To'rt nafar o'g'il bola va to'rt nafar qiz bolani ketma-ket joylashtirilgan 8 ta stulda o'tirishlari kerak. Bunda o'g'il bolalar juft o'rindagi, qiz bolalar esa toq o'rindagi stullarga o'tiradi. Buni necha usulda amalga oshirish mumkin?





MATEMATIKA

Yopiq turdagi (1–20) test topshiriqlari.

1. [0,9 ball] Hisoblang: $\frac{1-3+5-7+9-\dots-2027}{1-2+3-4+5-\dots-2028}$
A) -0,5 B) -1 C) 1 D) 2

2. [0,9 ball] n butun sonning nechta qiymatida $\frac{n^2-3}{n-2}$ ifodaning qiymati butun son bo'ladi?
A) 1 B) 2 C) 3 D) 4

3. [0,9 ball] Qavariq ko'pburchakning to'g'ri burchaklari eng ko'pi bilan nechta bo'lishi mumkin?
A) 1 B) 2 C) 3 D) 4

4. [0,9 ball] $22^2 + 44^2 + 66^2$ yig'indining eng katta tub bo'luvchisini toping.
A) 11 B) 7 C) 13 D) 17

5. [0,9 ball] Raqamlari yig'indisi 6 ga teng bo'lgan nechta 3 ga bo'linadigan ikki xonali son bor?
A) 6 B) 5 C) 4 D) 3

6. [0,9 ball] Berilgan tenglamani qanoatlantiruvchi x quyidagilardan qaysi biriga teng bo'lishi mumkin?
 $|x+6| = |x+10|$
A) -8 B) -6 C) 4 D) 6

7. [0,9 ball] 0 dan farqli a, b, c, d haqiqiy sonlar uchun $\frac{a}{b} = \frac{c}{d} = 6$ bo'lsa, $\frac{b}{a+b} \times \frac{d}{c-d}$ ko'paytmaning qiymatini toping.
A) 7 B) $\frac{1}{35}$ C) $\frac{1}{7}$ D) 35

8. [0,9 ball] 10 ga bo'lganda 9 qoldiq, 11 ga bo'lganda 10 qoldiq, 12 ga bo'lganda 11 qoldiq qoladigan eng kichik natural sonni toping.
A) 659 B) 329 C) 519 D) 689

9. [0,9 ball] Asoslari 12 va 8 ga teng bo'lgan teng yonli trapetsiyaning diagonallari o'zaro perpendikulyar bo'lsa, ushbu trapetsiyaning yuzini toping.

11. [1,5 ball] 60° li burchakka bir-biriga tashqi urinuvchi ikkita aylana ichki chizilgan. Agar ushbu aylanalardan kichigining radiusi 12 ga teng bo'lsa kattasining radiusini toping.
A) 18 B) 24 C) 36 D) 48

12. [1,5 ball] Berilgan $f(x) = \frac{2^{x+2} + 2^{1-x}}{2^{x+2} - x}$ funksiya uchun $f(t) = 3$ bo'lsa, $f(-t)$ ning qiymatini toping.
A) 2 B) 3 C) 4 D) 5

13. [1,5 ball] x_1 va x_2 sonlari $x^2 + 2x - 1 = 0$ tenglamaning yechimlari bo'lsa, $x_1^4 + x_2^4$ yig'indining qiymatini toping.
A) 34 B) 40 C) 1 D) 12

14. [1,5 ball] Quyidagi shartni qanoatlantiruvchi barcha musbat butun n sonlarning yig'indisini toping.
 $\frac{n-1}{n^2} + \frac{n-2}{n^2} + \dots + \frac{1}{n^2} = \frac{45}{1+3+5+\dots+2n-1}$
A) 9 B) 10 C) 12 D) 15

15. [1,5 ball] 0 dan farqli a, b, c, d haqiqiy sonlar uchun $\frac{a+b+c}{d} = \frac{a+b+d}{c} = \frac{a+c+d}{b} = \frac{b+c+d}{a} = r$ bo'lsa r ning qabul qilishi mumkin bo'lgan barcha qiymatlari yig'indisini toping.
A) 4 B) 3 C) 2 D) 1

16. [1,5 ball] Ixtiyoriy n natural son uchun $f(2n) = nf(n)$ shartni qanoatlantiradigan funksiya berilgan. Agar $f(1) = 1$ bo'lsa, $f(2^{10})$ ning qiymatini toping.
A) 2^{45} B) 2^{55} C) 2^{50} D) 2

17. [1,5 ball] Barcha ikki xonali natural sonlar ko'paytmasi 3^n soniga bo'linsa n natural sonning eng katta qiymatini toping.
A) 48 B) 41 C) 44 D) 40

18. [1,5 ball] 3^{2025} sonining oxirgi ikkita raqamini toping.



A) 150 B) 75 C) 100 D) 200

10. [0,9 ball] Ifodaning qiymatini toping:

$$\left(4\sqrt{4+2\sqrt{3}} - \sqrt{49+8\sqrt{3}}\right)^2$$

A) $3\sqrt{3}$ B) 6 C) $4\sqrt{3}$ D) 9

A) 49 B) 43 C) 27 D) 03

19. [1,5 ball] $p^2 - 2q^2 = 1$ tenglamani qanoatlantiradigan p va q tub sonlar uchun nechta $(p; q)$ juftliklar mavjud?

A) 1 B) 3 C) 2 D) 4

~ 2 ~

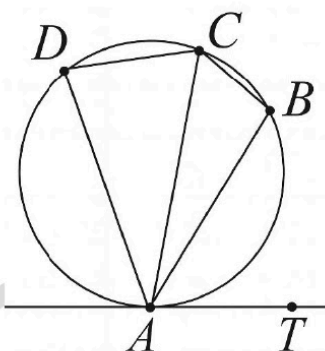
MATEMATIKA

20. [1,5 ball] Agar $\frac{a}{b} = \frac{b}{c} = \frac{c}{d}$, $a + d = 12$ va $ad = 11$ shartlar o'rinli bo'lsa, $b^3 + c^3$ yig'indining qiymatini toping.

A) $6\sqrt[3]{11}$ B) $6\sqrt{11}$ C) 66 D) 132

Ochiq turdagi (21–30) test topshiriqlari.

21. [2,6 ball] Berilgan rasmda $\angle BAT$, $\angle BAC$ va $\angle CAD$ burchaklarning nisbati mos ravishda 6: 2: 3 kabi. Agar AT to'g'ri chiziq berilgan aylanaga A nuqtada urinsa va $\angle ACD = 81^\circ$ bo'lsa, $\angle ABC$ necha gradusga teng?



Javob: _____

22. [2,6 ball] Berilgan yig'indining oxirgi raqamini toping:

$$1 + 4 \times 2 + 7 \times 2^2 + 10 \times 2^3 + \dots + 67 \times 2^{22}$$

Javob: _____

23. [2,6 ball] $\overline{BAA2025AAB}$ soni 7 ga bo'linsa, $A + B$ yig'indining eng kichik qiymatini toping.



Javob: _____

24. [2,6 ball] $xyz = 12^4$ tenglikni qanoatlantiradigan nechta $(x; y; z)$ natural sonlardan iborat uchliklar



mavjud? ((1; 2; 3) va (1; 3; 2) ikki xil hisoblanadi.)

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Javob: _____

25. [2,6 ball] Agar $\operatorname{tg} \alpha + \sin \alpha = 1$ va $(0 < \alpha < \frac{\pi}{2})$ bo'lsa, $(\sin 2\alpha + 2)^2$ ni hisoblang.

Javob: _____

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26. [2,6 ball] Raqamlari ko'paytmasi 10 dan katta bo'lmagan, lekin raqamlari yig'indisi 10 dan katta bo'lgan nechta uch xonali son mavjud?

Javob: _____

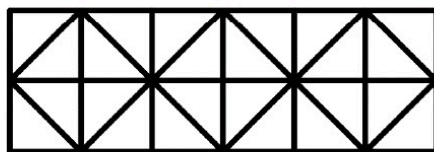
27. [2,6 ball] $ABCD$ trapetsiyada $\angle A = 90^\circ$, $\angle D = 30^\circ$. Markazi AD asosda bo'lgan aylana AB , BC va CD tomonlarga urinadi. Agar trapetsiyaning o'rta chizig'i $6 - \sqrt{3}$ ga teng bo'lsa, aylana radiusini toping.

Javob: _____

28. [2,6 ball] $\overline{abcd} \times 4 = \overline{dcba}$ shartni qanoatlantiruvchi $\overline{abcd}$ to'rt xonali sonning raqamlari yig'indisini toping.

Javob: _____

29. [2,6 ball] Berilgan rasmda nechta uchburchak mavjud?



Javob: _____

30. [2,6 ball] M nuqta $ABCD$ kvadratning ichida olingan. $MA = 1$, $MB = 2$, va $MC = 3$ bo'lsa, $\angle AMB$ necha gradusga teng?



Javob: _____

1. [0,9 ball] Hisoblang:

$$2025 \cdot 2024 \cdot 2023 \cdot \left(\frac{1}{4} \cdot \frac{2}{5} \cdot \frac{3}{6} \cdot \dots \cdot \frac{2022}{2025} \right).$$

A) 1 B) 2025 C) 6 D) 2022!

2. [0,9 ball] $1 \leq n \leq 20$ shartni

qanoatlantiradigan n natural son uchun $a = 13 \cdot 13 + n$ soni tub son bo'ladigan nechta n natural son mavjud?

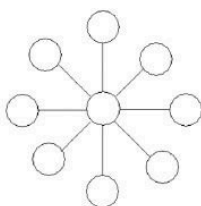
A) 2 B) 3 C) 4 D) 5

3. [0,9 ball] ABC teng yonli uchburchakda $AB = AC$ shart bajariladi. Agar BC asosda tanlangan D nuqta uchun $AD = CD$ va $\angle BAD = 36^\circ$ tengliklar o'rinli bo'lsa, $\angle ABD$ necha gradusga teng?

A) 36 B) 42 C) 48 D) 54

4. [0,9 ball] 2 dan 10 gacha bo'lgan natural sonlar (har bir son faqat bir marta ishlatiladi) quyidagi shakldagi to'qqizta doiraga joylashtirilgan. Bir to'g'ri chiziqda yotuvchi 3 ta doiracha ichidagi sonlarning yig'indilari teng bo'lsa, ushbu yig'indi necha xil qiymat qabul qila oladi?

A) 3 B) 4 C) 5 D) 6



5. [0,9 ball] Berilgan ketma-ketlik

1, 2, 3, 4, 5, 6, 7, 6, 5, 4, 3, 2, 1, 2, 3, 4, 5, 6, 7, 6, 5, ... ko'rinishda bo'lib, cheksiz marta 1 dan 7 gacha o'sib, 7 dan 1 gacha kamayadi. Ushbu ketma-ketlikning 2025-hadini toping.

A) 1 B) 7 C) 6 D) 5

6. [0,9 ball] $\triangle ABC$ da $\angle BAC = 25^\circ$ va $\angle BCA > 90^\circ$. Aytaylik B_1 nuqta B nuqtaning AC tog'ri chiziqqa nisbatan simmetrigi, C_1 nuqta C nuqtaning AB to'g'ri chiziqqa nisbatan simmetrigi bo'lsin. Agar $\angle B_1BC_1 = 100^\circ$ bo'lsa, $\angle BCA$ necha gradusga teng?

A) 110° B) 120° C) 130° D) 140°

7. [0,9 ball] Dastlabki 100 ta natural sondan nechtasini $2a + 3b$ ko'rinishida ifodalab bo'lmaydi? (Bunda a va b nomanfiy butun sonlar.)

A) 3 B) 5 C) 1 D) 4

8. [0,9 ball] n natural son uchun

$$n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n \text{ bo'lsa,}$$

$1! + 2! + 3! + \dots + 2025!$ yig'indining oxirgi ikkita raqamining yig'indisini toping.

A) 3 B) 13 C) 4 D) 6

9. [0,9 ball] a va b natural sonlar uchun

$4a - 7b + 28ab = 2020$ bo'lsa, ab ko'paytma quyidagilardan qaysi biriga teng bo'lishi mumkin?

A) 96 B) 102 C) 106 D) 108

10. [0,9 ball] Agar $f(x) = x^2 + kx + 110$ va $f(1) = f(20)$ bo'lsa, $f(10)$ ning qiymatini toping.

A) -21 B) 0 C) 100 D) 420

11. [1,5 ball] Eng katta n natural sonni toping bunda: $n!$ soni 7^{72} ga bo'linmasin. Bunda $n! = 1 \cdot 2 \cdot 3 \cdot \dots \cdot n$

A) 433 B) 440 C) 439 D) 441

12. [1,5 ball] Jahongir do'konga kirib, 10 dollarlik mahsulotni 10 % chegirma bilan, 15 dollarlik mahsulotni 15 % chegirma bilan va 25 dollarlik mahsulotni 25 % chegirma bilan sotib oldi. Jahongir do'kondagi xaridida umumiy necha foiz chegirma oldi?

A) 19 % B) $\frac{50}{3}$ % C) 50 % D) 20 %

13. [1,5 ball] Tenglamaning barcha ildizlari yig'indisini toping:

$$3 \cdot |x^2 - (2 - x)^2 + 1| = 10 - |8x - 6|$$

A) 2 B) $\frac{5}{4}$ C) 1 D) $\frac{3}{2}$

14. [1,5 ball] ABC uchburchakda $\angle ACB = 90^\circ$ va AL bissektrisa, CM mediana bo'lsin. ($L \in BC$ va $M \in AB$) Agar $AL \perp CM$ bo'lsa, $\angle ABC$ necha gradusga teng?

A) 60° B) 15° C) 45° D) 30°

15. [1,5 ball] O'nlik va birlik raqamlari o'rin almashtirilsa, hosil bo'lgan yangi son bilan asl son o'zaro tub bo'ladigan 10 dan katta va 40 dan kichik nechta natural son mavjud?

A) 12 B) 14 C) 16 D) 10

16. [1,5 ball] ABC uchburchakda

$\angle A : \angle B : \angle C = 2 : 3 : 7$ va BD kesma mediana bo'lsa, $\angle ABD$ ning qiymatini toping. (D nuqta AC kesmada yotadi.)

A) 10° B) 15° C) $22,5^\circ$ D) 30°

17. [1,5 ball] a, b, c haqiqiy sonlar uchun $\frac{a}{b} = \frac{b}{c} = k$ va $\frac{a+3b-18c}{b+6c} = \frac{5}{3}$ bo'lsa, k sonning qiymatini toping.

A) -6 B) $\frac{7}{3}$ C) $\frac{14}{3}$ D) $\frac{18}{5}$

18. [1,5 ball] p, q, r tub sonlar uchun $pqr + 2025 = 3(pq + qr + pr)$ bo'lsa, $p + q + r$ yig'indining qiymatini toping.

A) 220 B) 227 C) 225 D) 228

19. [1,5 ball] Eng katta n natural sonni toping bunda $n^5 - 2100$ soni

$(n^2 - n + 1)^2 + n(n - 1)^2$ soniga bo'linsin.

A) 2 B) 11 C) 7 D) 5

20. [1,5 ball] Berilgan

3×3 jadvalda har bir qatordagi sonlar yig'indisi teng va har bir ustundagi sonlar yig'indisi teng (qator va ustundagi sonlar yig'indisi teng bo'lishi shart emas.) Agar berilgan rasmda $a + b = 18$ bo'lsa, x ning qiymatini toping.

A) 7 B) 8 C) 9 D) 10

5		x
a	7	
	b	11

21. [2,6 ball] 10 000 dan kichik bo'lgan nechta natural sonning tarkibida aynan ikkita "2" raqami va aynan bitta "5" raqami bor? (masalan, 2025).

A) 96 B) 93 C) 84 D) 99

22. [2,6 ball] $a < b < c$ butun sonlar quyidagi shartlarni qanoatlantiradi: b soni a va c sonlarning o'rta arifmetigi, c soni a va b sonlarning o'rta geometrigi. abc ko'paytmaning eng kichik qiymatini toping.

A) 12 B) 8 C) 6 D) -12

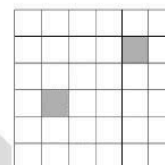
23. [2,6 ball] $f(n)$ soni n ta natural bo'luvchiga ega eng kichik natural son bo'lsin. Masalan: $f(2) = 2$, $f(3) = 4$ bo'lsa, $f(8) + f(9)$ yig'indining qiymatini toping.

A) 384 B) 54 C) 260 D) 60

24. [2,6 ball] $ABCD$ to'rtburchak tomonlarining o'rtalari mos ravishda M, N, P, Q nuqtalar bo'lsin (M — AB ning o'rtasi, N — BC ning o'rtasi, P — CD ning o'rtasi, Q — DA ning o'rtasi). AC diagonalda S nuqta shunday tanlanganki bunda $QSPD$, $AMSQ$ va $MBNS$ to'rtburchaklarning yuzalari mos ravishda 22, 24 va 32 ga teng. $AS : SC$ nisbat nechaga teng?

A) $\frac{4}{5}$ B) $\frac{8}{11}$ C) $\frac{3}{8}$ D) $\frac{6}{11}$

25. [2,6 ball] Berilgan rasmda nechta to'g'ri to'rtburchak aynan bitta bo'yalgan kvadratchani o'z ichiga oladi?



A) 164 B) 417 C) 196 D) 172

26. [2,6 ball] $a_5 = 5$ va har bir $n \geq 6$ natural son uchun $a_n = 1000a_{n-1} + n$ tenglik o'rinli. $n \geq 6$ da a_n soni 111 ga bo'linadigan n natural sonlarning dastlabki ikkitasi yig'indisining qiymatini toping.

A) 221 B) 110 C) 341 D) 119

27. [2,6 ball] k haqiqiy sonni toping bunda: $x^3 - 30x^2 + kx - 840 = 0$ tenglamaning ildizlari arifmetik progressiya tashkil qilsin.

A) 300 B) 284 C) 840 D) 256

28. [2,6 ball] Ikkita beshburchakning jami 10 ta uchi ranglar bilan bo'yalgan: 3 tasi qizil, 4 tasi oq va 3 tasi ko'k. Hech bir beshburchakning tomoni ikkita qizil yoki ikkita ko'k uchni tutashtirmaslik ehtimoli $\frac{m}{n}$ ga teng bo'lsa, $m + n$ yig'indini hisoblang. (bu yerda m va n o'zaro tub natural sonlar.)

A) 101 B) 97 C) 93 D) 85

29. [2,6 ball] a va b har xil haqiqiy sonlar uchun $2a^3 + (1 + \sqrt{3})ab + 2b^3 = \frac{(5+3\sqrt{3})}{54}$ bo'lsa, $(6a + 6b - 1)^6$ ning qiymatini toping.

A) 24 B) 27 C) 36 D) 18

1-qism: Har bir topshiriq 0,9 balldan baholanadi

- Ifodani soddalashtiring: $\frac{1-8b}{\sqrt{2+4b}} - \frac{1-8b}{\sqrt{2-4b}}$ A) $\sqrt{2}$ B) $4b$ C) $-4b$ D) $\sqrt{2} - 4b$
- $A(m; n)$ nuqta $y + 1 = 2x^2 - 4x$ parabola uchi bo'lsa, $m - n$ ayirmani toping.
A) 1 B) 2 C) 3 D) 4
- Hisoblang: $tg \frac{2\pi}{3} \cdot \sin\left(-\frac{3\pi}{4}\right) \cdot \cos \frac{\pi}{6}$ A) $-\frac{3\sqrt{2}}{4}$ B) $-\frac{\sqrt{6}}{4}$ C) $\frac{\sqrt{6}}{4}$ D) $\frac{3\sqrt{2}}{4}$
- Tengsizlikni yeching: $5x - 20 \leq x^2 \leq 8x$
A) $[0; 10]$ B) $[0; 8]$ C) $[0; 2] \cup [8; 10]$ D) $[0; 1] \cup [8; 10]$
- 9 ta avtobus va 12 ta tramvay 966 nafar yo'lovchini tashiydi. 4 ta avtobus va 17 ta tramvay esa 931 nafar yo'lovchi tashiydi. Bitta avtobus va bitta tramvayga sig'adigan yo'lovchilar soni bir-biridan qancha farq qiladi? A) 6 B) 5 C) 7 D) 8
- $y = \frac{12}{\sin x + 5}$ funksiyaning qiymatlari sohasini toping
A) $[0; 2]$ B) $[2; 3]$ C) $[2; 2,4]$ D) $[2,4; 3]$
- Geometrik progressiyaning birinchi, uchinchi va o'n birinchi hadlari ko'paytmasi 8 ga teng bo'lsa, progressiyaning ikkinchi va sakkizinchi hadlari ko'paytmasini toping.
A) 16 B) 2 C) 4 D) 8
- $tg \alpha = \sqrt{11}$ bo'lsa, $\sin 2\alpha$ ni toping. A) $\frac{\sqrt{11}}{12}$ B) $\frac{\sqrt{11}}{4}$ C) $\frac{\sqrt{11}}{11}$ D) $\frac{\sqrt{11}}{6}$
- Markazi O nuqtada bo'lgan aylananing AB va AC vatarlar o'zaro perpendikulyar. Agar AOC burchak 100° bo'lsa, AOB burchakni toping.
A) 80° B) 100° C) 50° D) 90°
- Ifodani soddalashtiring: $\frac{\cos 2\alpha}{\sin \alpha \cos \alpha + \sin^2 \alpha} + 1$ A) $tg \alpha$ B) $ctg \alpha$ C) $\cos \alpha$ D) $\sin \alpha$

2-qism: Har bir topshiriq 1,5 balldan baholanadi

- $\frac{\sqrt{3}-\sqrt{x-4}}{\sqrt{8-x}} > 0$ tengsizlikning butun yechimlari o'rta arifmetigini toping.
A) 4,5 B) 5 C) 5,5 D) 6
- ABC uchburchakning AB tomoniga parallel to'g'ri chiziq AC va BC tomonlarni mos ravishda K va N nuqtalarda kesib o'tadi. Agar $\triangle KCN$ uchburchak yuzi $\triangle AKNB$ trapetsiya yuzidan ikki marta katta bo'lsa, $AK : CK$ nisbatni toping.
A) $(2 - \sqrt{2}) : \sqrt{2}$ B) $(2 - \sqrt{3}) : \sqrt{2}$ C) $(\sqrt{3} - 1) : 1$ D) $(\sqrt{3} - \sqrt{2}) : \sqrt{2}$
- Agar $(a - b)^2 + (b - c)^2 + (c - a)^2 = 6$ va $(a + b)^2 + (b + c)^2 + (c + a)^2 = 18$ bo'lsa, $(a + b)(b + c) + (b + c)(c + a) + (c + a)(a + b)$ ni toping.
A) 15 B) 24 C) 12 D) 20
- Arifmetik progressiyada $a_1 + a_7 = 12$, $a_3 \cdot a_5 = 28$ bo'lsa, $a_3^2 + a_5^2$ yig'indini toping.
A) 56 B) 116 C) 92 D) 88
- $x_1 = p + 2q$, $x_2 = 4p + 7q$ ($p \neq 0$) sonlar $x^2 + px + q = 0$ tenglama ildizlari bo'lsa, $p + q$ yig'indini toping. A) -1 B) -2 C) 2 D) 3



16. Burchaklaridan biri 120° , yuzi esa $\sqrt{3}$ ga teng bo'lgan tengyonli uchburchakka tashqi chizilgan aylana radiusini toping. A) $\frac{1}{\sqrt{3}}$ B) 1 C) 2 D) $\sqrt{3}$
17. Ikki sonni bir biriga bo'lganda bo'linma ham, qoldiq ham 4 ga teng. Bo'linuvchi va bo'luvchining ayirmasiga, ularning o'rta arifmetigini, bo'linmani va qoldiqni qo'shsak 47 hosil bo'ladi. Bo'linuvchini toping. A) 28 B) 32 C) 24 D) 26
18. $f(x)$ funksiya uchun $f(0) \cdot (f(x) + 2) = 4x - 1$ tenglik o'rinli bo'lsa, $f(-1)$ ni toping. A) 2 B) -1 C) -5 D) 3
19. $1 \cdot 2 \cdot 3; 2 \cdot 3 \cdot 4; 3 \cdot 4 \cdot 5; 4 \cdot 5 \cdot 6; \dots; 98 \cdot 99 \cdot 100$ ko'paytmalardan nechitasi 12 ga bo'linadi? A) 72 B) 73 C) 74 D) 75
20. Tenglamaning haqiqiy ildizlari sonini toping. $(x - 1)|x| = |x| + 1$
A) 1 B) 2 C) 3 D) $\emptyset$

3-qism: Har bir topshiriq 2,6 balldan baholanadi

21. a, b, c musbat sonlar $ab = 6$ va $bc = 15$ tengliklarni qanoatlantiradi. $a + b + 2c$ ifodaning eng kichik qiymatini toping.
22. Teng yonli trapetsiyaga ichki chizilgan aylana radiusi 3 ga teng. Agar trapetsiyaning asoslaridan biri balandligidan 2 marta uzun bo'lsa, trapetsiya yuzini toping.
23. $[2u] + 2\{u\} = 20,24$ tenglamaning ildizlari yig'indisini toping, bunda $[u]$ – u sonining butun qismi, $\{u\}$ – u sonining kasr qismi.
24. x va y sonlari $x(1 - y) = 2y(x + 1) = 2$ tenglikni qanoatlantiradi. $(x + x^2)(y - y^2)(x + y)$ ifodaning qiymatini toping
25. ABCD qavariq to'rtburchakning AC diagonalini o'rtasidan barcha tomonlarga masofalar bir xil 4,8 ga teng. To'rtburchakning BD diagonalini 12 ga teng bo'lsa, to'rtburchak yuzini toping.
26. $\frac{x^2 - 2px + p^2 + 8p + 8}{x - 1} = 0$ tenglama yagona yechimga ega bo'ladigan p parametrning qiymatlari yig'indisini toping.
27. $\frac{35 + 2n}{3n + 1}$ ifoda butun son bo'ladigan barcha n natural sonlar nechta?
28. $\begin{cases} 2x^2 + 3y = -8,5 \\ z^2 + 4x = -1 \\ y^2 + 2z = 7 \end{cases}$ tenglamalar sistemasidan $2x + 3y + 4z$ ni toping.
29. Yuzduzchalarni raqamlarga almashtirganda $**253*$ ko'rinishdagi nechta son 72 ga karrali bo'ladi?
30. To'g'ri burchakli uchburchakning to'g'ri burchagidan chiquvchi balandlik 6 ga, katetlarning gipotenuzadagi proyeksiyalarining farqi esa 5 ga teng. Berilgan uchburchak yuzini toping.



Yopiq turdagi (1–20) test topshiriqlari.

1. [0,9 ball] $2^2 + 3^3 + 4^4 + 5^5 + 6^6$

yig'indining oxirgi raqamini toping.

A) 8 B) 4 C) 2 D) 0

2. [0,9 ball] $0,9 < x < 9,5$ shartni qanoatlantiruvchi barcha natural x larning yig'indisining turli tub bo'luvchilari sonini toping.

A) 1 B) 2 C) 3 D) 4

3. [0,9 ball] x , 2 , $x + 2y$ va 4 sonlari arifmetik progressiyani tashkil qilsa, $x^2 + y^2$ ning qiymatini toping.

A) 1 B) 2 C) 3 D) 4

4. [0,9 ball] $16^a \times 9^a = 6^b \times 8^2$ tenglikdan foydalanib a va b natural sonlarning yig'indisini hisoblang.

A) 9 B) 15 C) 12 D) 6

5. [0,9 ball] Agar x va y musbat sonlar $\log_3(x^2 y^2) = (2 \log_3 y) + 6$ tenglikni qanoatlantirsa, x ning qiymatini toping.

A) 1 B) 4 C) 79 D) 27

6. [0,9 ball] Natural bo'luvchilari soni aynan 3 ta bo'lgan dastlabki 3 ta natural sonning yig'indisini toping.

A) 36 B) 38 C) 40 D) 42

7. [0,9 ball] Agar x_1 va x_2 sonlar $2x^2 - 5x + 3 = 0$ tenglamaning ildizlari bo'lsa, $x_2 + x_1 \times x_2 + x_1$ ifodaning qiymatini toping.

A) 4 B) 5 C) 6 D) 7

8. [0,9 ball] p tub son berilgan. $24p$ sonining natural bo'luvchilari soni eng ko'pi bilan nechta bo'lishi mumkin?

A) 10 B) 16 C) 14 D) 12

9. [0,9 ball] Berilgan tengsizlikni qanoatlantiruvchi barcha butun sonlar yig'indisini toping.

$$(x - 8)(8x^2 + 8)(8x + 8) \leq 0$$

A) 27 B) 35 C) 36 D) 44

10. [0,9 ball] OY o'qi, $y = x$ va $y = -x + 6$ to'g'ri chiziqlar bilan chegaralangan uchburchak yuzini toping.

A) 36 B) 18 C) 9 D) 4,5

11. [1,5 ball] $2025!$ sonining $2024!$ sonidan katta bo'luvchilari soni nechta?

A) 1 B) 2025 C) 14 D) 2024

12. [1,5 ball] Agar

A) 1 B) 2 C) 3 D) 4

13. [1,5 ball] O'suvchi arifmetik progressiyaning dastlabki 10 ta hadining yig'indisi 140 ga teng bo'lib, ikkinchi va to'qqizinchi hadlari ko'paytmasi 147 ga teng bo'lsa, uchinchi hadni toping?

A) 3 B) 5 C) 7 D) 9

14. [1,5 ball] Trapetsiyaning o'rta chizig'i 14 ga teng va katta asosidagi burchaklari 30° va 60° ga teng. Asoslari o'rtalarini tutashiruvchi kesma uzunligi 8 ga teng bo'lsa, katta asosining uzunligini toping.

A) 18 B) 20 C) 22 D) 24

15. [1,5 ball] 4 nafar turli yoshdagi kishilarning ixtiyoriy ikkitasining yoshlari farqi turlicha ekanligi ma'lum. Ularning eng kattasi va eng kichigining yoshlari farqi eng kamida nechga teng bo'lishi mumkin?

A) 7 B) 5 C) 8 D) 6

16. [1,5 ball] n natural son uchun ushbu $\frac{n^3 - 2n^2 - 12}{n}$ nisbatning barcha natural qiymatlari yig'indisini toping.

A) 27 B) 143 C) 124 D) 146

17. [1,5 ball] $ABCD$ to'rtburchak diagonallari orqali 4 ta uchburchakka ajratilgan. Ushbu 4 ta uchburchakdan faqat bitta umumiy uchga ega bo'lgan uchburchaklarning yuzalari 16 va 24 ga teng va uchinchi uchburchakning yuzi 12 ga teng bo'lsa, oxirgi uchburchakning yuzini toping.

A) 28 B) 32 C) 18 D) 20

18. [1,5 ball] $1^2, 2^2, 3^2, 4^2, 5^2, 6^2, 7^2$ va 8^2 uzunlikdagi kesmalardan foydalanib nechta turli tomonli uchburchak yasash mumkin?

A) 4 B) 49 C) 6 D) 8

19. [1,5 ball] Hozir soat 5:20 bo'lsa, necha daqiqadan so'ng soat va minut mili birinchi marta ustma-ust tushadi?

A) $\frac{80}{11}$ B) 7 C) $\frac{15}{2}$ D) $\frac{25}{3}$

20. [1,5 ball] $ABCD$ parallelogram ichida P nuqta shunday tanlanganki, bunda $\angle APB + \angle CPD = 180^\circ$ va $\angle PBC = 36^\circ$. $\angle PDC$ necha gradusga teng?

A) 18 B) 54 C) 72 D) 36

 $f(x) = \frac{1}{1+2^{\lg x}} + \frac{1}{1+4^{\lg x}} + \frac{1}{1+8^{\lg x}}$ bo'lsa,
 $f(x) + f\left(\frac{1}{x}\right)$ yig'indining qiymatini toping.

~ 2 ~

MATEMATIKA

Ochiq turdagi (21–30) test topshiriqlari.

21. [2,6 ball] $(x^3 - 2025x + 1)^3$ ko'phad ochib chiqilganda x^6 oldidagi koeffitsiyentini toping.

Javob: _____

22. [2,6 ball] Berilgan tengsizlikni qanoatlantiruvchi 7 ta butun x bo'lsa, a natural sonning qiymatini toping.

$$6x^2 - ax - a^2 < 0$$

Javob: _____

23. [2,6 ball] Teng yonli ABC uchburchak berilgan ($AC = BC$). Ushbu uchburchakning BD bissektisasi o'tkazilgan ($D \in AC$). BD to'g'ri chiziqqa D nuqta orqali o'tkazilgan perpendikulyar chiziq BA to'g'ri chiziqni M nuqtada kesib o'tadi. Agar $AD = 12$ bo'lsa, BM kesmaning uzunligini toping.

Javob: _____

24. [2,6 ball] Agar $F(n) = n^5 + 3n^3 - 4n$ bo'lsa, $EKUB(F(2); F(3); F(4); \dots F(20))$ ifodani hisoblang.

Javob: _____

25. [2,6 ball] 11222333344444 ... ushbu ketma-ketlikda 2 ta 1, 3 ta 2, ... $n + 1$ ta n soni bitta qatorda yozilgan. Ketma-ketlikdagi 2025-o'rindagi raqamni toping.

Javob: _____

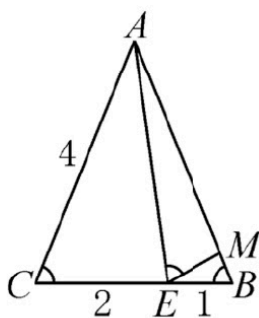
26. [2,6 ball] a va b natural sonlar bo'lib, $13a + 31b$ ko'rinishida kamida 3 xil usulda yozish mumkin bo'lgan eng kichik sonni toping.

Javob: _____



MATEMATIKA

27. [2,6 ball] ABC teng yonli uchburchakning BC asosida E nuqta va AB yon tomonida M nuqta $\angle B = \angle C = \angle AEM$ shart o'rinli bo'ladigan qilib tanlangan. Agar $BE = 1$, $CE = 2$ va $AC = 4$ tengliklar o'rinli bo'lsa, MA kesmaning uzunligini toping.



Javob: _____

28. [2,6 ball] Dastlabki 8888 ta natural sonning nechtasida 8 raqami mavjud?

Javob: _____

29. [2,6 ball] $20x^{11} + bx^{10} + cx^9 + 4$ ko'phad $x^2 - x + 1$ ko'phadga qoldiqsiz bo'linsa, bc ko'paytmaning qiymatni toping.

Javob: _____

30. [2,6 ball] $\{1, 2, 3, \dots, 10\}$ to'planning nechta shunday qism to'plami mavjud bunda: ushbu qism to'planning hech qaysi ikkita elementi ketma-ket emas?

Javob: _____

1. [0,9 ball] Quyidagi qator yig'indisini hisoblang:

$$\sum_{k=0}^{2025} \frac{2025! \cdot (-1)^k 2^k}{k! \cdot (2025 - k)!}$$

A) $\frac{1-(-2)^{2025}}{3}$ B) -1 C) 1 D) 3^{2025}

2. [0,9 ball] Agar x va y sonlar $\frac{x-y}{x+y} = 7$ va $\frac{xy}{x+y} = -84$ shartlarni qanoatlantirsa, $(x+y) + (x-y) + xy$ ifodaning qiymati topilsin.

A) -532 B) -568 C) -588 D) -600

3. [0,9 ball] $\{a\}$ - a sonining kasr qismi bo'lsa, quyidagi yig'indini hisoblang:

$$\{2 - \sqrt{2}\} + \{2 + \sqrt{2}\} + \{3 - \sqrt{3}\} + \{3 + \sqrt{3}\} + \{4 - \sqrt{4}\} + \{4 + \sqrt{4}\} + \dots + \{2025 - \sqrt{2025}\} + \{2025 + \sqrt{2025}\}$$

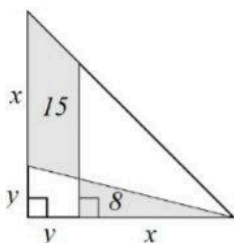
A) 1979 B) 1980 C) 2024 D) 2025

4. [0,9 ball] Haqiqiy musbat a son uchun $f(x) = \frac{1}{a} - \frac{1}{a^{x+1}}$ funksiya toq funksiya bo'lsa, ushbu funksiyaning qiymatlar to'plamini aniqlang.

A) $(-1; 1)$ B) $(-\infty; 1)$

C) $(-\frac{1}{2}; \frac{1}{2})$ D) $(-\infty; 0)$

5. [0,9 ball] To'g'ri burchakli uchburchakda ikkita bo'yalgan yuza maydonlari 15 va 8 ga teng. x va y kesmalar chizmada ko'rsatilgan. xy ko'paytmani toping.



A) $2\sqrt{30}$ B) 14 C) 11,5 D) 12

6. [0,9 ball] 12 ta o'quvchini 4 kishilik uchta guruhga necha xil usulda ajratish mumkin.

A) 5760 B) 5775 C) 34650 D) 40320

7. [0,9 ball] r haqiqiy son uchun $\sqrt[3]{r} - \frac{1}{\sqrt[3]{r}} = 1$ bo'lsa, $r^3 - \frac{1}{r^3}$ ning qiymatini toping.

A) 76 B) -2 C) 33 D) 36

8. [0,9 ball] 10 ta qizil va 10 ta ko'k shar berilgan. Sharlarni bir qatorga shunday joylashtirish kerakki, hech qanday ikkita qizil shar yonma-yon joylashmasin. Joylashtirish usullari sonini toping.

A) 1 B) 11 C) 20 D) 21

9. [0,9 ball] $(1 + \sin \alpha)^{\frac{1}{3}} + (1 - \sin \alpha)^{\frac{1}{3}} = 1,5$ bo'lsa, $(1 + \sin \alpha)^{\frac{2}{3}} - (\cos \alpha)^{\frac{2}{3}} + (1 - \sin \alpha)^{\frac{2}{3}}$ ning qiymatini toping.

A) $\frac{2}{3}$ B) $\frac{4}{3}$ C) $\frac{3}{2}$ D) 3

10. [0,9 ball] 20^{22} sonining to'la kvadrat yoki to'la kub bo'luvchilari nechta?

A) 345 B) 396 C) 364 D) 342

11. [1,5 ball] Agar har bir talaba kamida bitta kitob olsa, 5 ta turli kitobni 3 ta talabaga necha usulda taqsimlash mumkin?

A) 144 B) 150 C) 240 D) 243

12. [1,5 ball] Quyidagi

$$3x^2y - 12xy - 8y - 7 = 0$$

tenglamani qanoatlantiruvchi nechta butun $(x; y)$ juftliklar mavjud?

A) 0 B) 1 C) 2 D) 4

13. [1,5 ball] f va g funksiyalar quyidagi jadval bilan aniqlangan, unda ba'zi qiymatlar savol belgisi (?) bilan ko'rsatilgan. Jadvalda shuningdek, $h(x) = g(f(x))$ tenglik bilan aniqlangan h funksiyasining ba'zi qiymatlari mavjud bo'lib, ular ham savol belgisi (?) bilan almashtirilgan.

x	0	1	2	3	4	5
$f(x)$	5	?	?	7	1	2
$g(x)$	?	7	?	0	10	8
$h(x)$	?	8	0	5	?	9

$g(2) + h(4)$ ni hisoblang.

A) 5 B) 8 C) 15 D) 16

14. [1,5 ball] p va q funksiyalar uchun $p(x) = q(30x) + 33$ tenglik o'rinli. Agar $q(5) = 16$ va $p(5) = 629$ bo'lib, q – chiziqli funksiya bo'lsa, $p(x)$ funksiyaning formulasini toping.

A) $p(x) = 120x + 29$

B) $p(x) = 120x - 87$

C) $p(x) = 30x^2 - 60x + 179$

D) $p(x) = 36x^2 + 10x - 321$

15. [1,5 ball] Agar $x^2 + y^2 = 2x + 6y + 6$ bo'lsa, $4x + 3y$ ifodaning eng katta qiymatini toping.

A) 33 B) 32 C) 74 D) 73

16. [1,5 ball] Agar m va n o'zaro tub sonlar bo'lsa, $EKUB(30m + n; 15n + 2m)$ eng ko'pi bilan qaysi qiymatni qabul qilishi mumkin?

A) 448 B) 224 C) 450 D) 225

17. [1,5 ball] $m^2 + 2n^2 = 3(m + n)$ tenglama butun sonlarda nechta juft $(m; n)$ yechimga ega?

A) 0 B) 2 C) 4 D) cheksiz ko'p

18. [1,5 ball] Stol ustida 100 ta tanga joylashgan bo'lib, ularning barchasining raqam

tomoni yuqoriga qaragan. Har bir yurishda aynan 7 ta tangani aylantirish mumkin (ya'ni har bir aylantirilgan tanga orqaga o'giriladi: raqam tomondan gerb tomonga yoki aksincha). Kamida necha yurishdan keyin barcha tangalarning gerb tomoni yuqoriga qaragan bo'lishi mumkin?

A) 15 B) 16 C) 18 D) barcha tangalarning gerb tomonini yuqoriga qaratish imkonsiz

19. [1,5 ball] Agar

$T = \left(1 + \frac{1}{5^2}\right) \left(1 + \frac{1}{5^4}\right) \left(1 + \frac{1}{5^8}\right) \left(1 + \frac{1}{5^{16}}\right) \cdot \dots$ bo'lsa, $\sqrt[5]{9^{12T}}$ ning qiymatini toping.

A) 9 B) 27 C) 81 D) 243

20. [1,5 ball] Jasur 5 dan 11 gacha bo'lgan oraliqdan haqiqiy sonni tasodifan tanlaydi, Temur esa 3 dan 10 gacha bo'lgan oraliqdan haqiqiy sonni tasodifan tanlaydi. Ularning tanlagan sonlari orasidagi farq ko'pi bilan 2 ga teng bo'lish ehtimolini toping.

A) $\frac{29}{42}$ B) $\frac{13}{28}$ C) $\frac{23}{42}$ D) $\frac{23}{28}$

21. [2,6 ball] $f: N^2 \rightarrow N$ funksiya uchun $f(2; 1) = 1$ bo'lib

$$f(m + 1; n) = f(m; n) + m$$

$$f(m; n + 1) = f(m; n) - n$$

tengliklar o'rinli. $f(p; q) = 2025$ tenglama nechta yechimga ega? Bu yerda N – natural sonlar to'plami.

A) 15 B) 20 C) 30 D) 45

22. [2,6 ball] Soha tengsizliklar bilan berilgan: $y \geq 3x^2$, $y \leq 6 - \frac{3}{2}x$, $x \geq 0$.

Bu sohada, bir tomoni y o'qida yotuvchi yuzi eng katta to'g'ri to'rtburchak chizilgan. To'g'ri to'rtburchakning yuzasini toping.

A) $\frac{22}{9}$ B) $\frac{9}{4}$ C) $\frac{20}{9}$ D) $\frac{7}{4}$

23. [2,6 ball] $\{a_k\}$ va $\{b_k\}$ ketma-ketliklar uchun $a_k = (k^2 + 1)k!$ va $b_k = a_1 + a_2 + a_3 + \dots + a_k$ bo'lsin.

$\frac{a_{2025}}{b_{2025}} = \frac{m}{n}$ bo'lib m va n o'zaro tub natural sonlar bo'lsa, $n - m$ ning qiymatini toping.

- A) 2025 B) 2024 C) 1012 D) 4050

24. [2,6 ball] $f: R \rightarrow R$ va $g: R \rightarrow R$ funksiyalar uchun

$$\sin x + \cos y = f(x) + f(y) + g(x) - g(y)$$

tenglik barcha $x, y \in R$ lar uchun o'rinli

bo'lib, $g\left(\frac{5\pi}{4}\right) = 1$ bo'lsa, $g(\pi)$ ni hisoblang.

- A) 1 B) $\frac{1}{2}$ C) $\frac{3}{2}$ D) $\frac{\sqrt{2}}{2}$

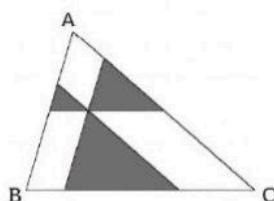
25. [2,6 ball] $x^2 - 8[x] + 12 = 0$ tenglama nechta ildizga ega. Bu yerda $[a] - a$ dan kichik eng katta butun son.

- A) 2 B) 3 C) 4 D) 5

26. [2,6 ball] Agar $x + y + z = 1$, $x^2 + y^2 + z^2 = 2$ va $x^3 + y^3 + z^3 = 3$ bo'lsa, $x^4 + y^4 + z^4$ ning qiymati topilsin.

- A) $\frac{13}{5}$ B) $\frac{25}{6}$ C) $\frac{19}{4}$ D) $\frac{29}{6}$

27. [2,6 ball] ABC uchburchak ichidagi biror nuqtadan asoslarga parallel to'g'ri chiziqlar o'tkazilganda yuzalari 2, 6 va 18 ga teng bo'lgan uchburchaklar hosil bo'ladi. (Chizma bo'yab ko'rsatilgan). U holda ABC uchburchak yuzini toping.



- A) $38 + 16\sqrt{3}$ B) $48 + 12\sqrt{3}$
C) $24 + 16\sqrt{3}$ D) $24 + 12\sqrt{3}$

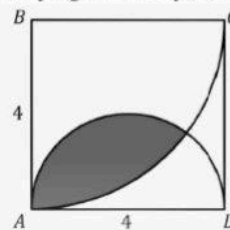
28. [2,6 ball] Faqat 1 dan 9 gacha bo'lgan raqamlardan foydalangan holda (0 raqami ishtirok etmaydi), 11 ga karrali nechta uch

xonali son tuzish mumkin?

(Raqamlar takrorlanishi mumkin.)

- A) 81 B) 70 C) 66 D) 64

29. [2,6 ball] ABCD kvadratda B ni markaz qilib radiusi 4 ga teng aylana va AD ni diametr qilib diametri 4 ga teng aylana chizilgan. Bo'yalgan shakl yuzasini toping.



- A) $8\pi - 8 - 16\arctg 2$
B) $8\pi - 4 - 16\arctg 2$
C) $8\pi - 8 - 12\arctg 2$
D) $16\pi - 8 - 12\arctg 2$

30. [2,6 ball] Jamolda 2 ta tanga, Hamidda 3 ta tanga bor. Ular 3 marta, har safar bittadan, tasodifiy tanga tanlab almashishdi. Har bir almashishda Jamol ham, Hamid ham qo'lidagi tangalardan birini teng ehtimollik bilan tanlaydi va ularni almashtiradi. 3 ta almashishdan keyin Jamol ham, Hamid ham o'zlarining boshlang'ich tangalarini qayta qo'lida saqlab qolish ehtimoli qancha?

- A) $\frac{1}{12}$ B) $\frac{1}{24}$ C) $\frac{1}{48}$ D) $\frac{5}{48}$

1-qism: Har bir topshiriq 0,9 ball dan baholanadi

1. Ifodani soddalashtiring: $2\sqrt{1,125} - 5\sqrt{1,62} + 3\sqrt{4,5}$
A) $0,75\sqrt{2}$ B) $1,5\sqrt{2}$ C) $0,5\sqrt{2}$ D) $3\sqrt{2}$
2. p va $p^4 + 3$ – sonlari tub sonlar bo'lsa, $p^3 + 3$ ni toping.
A) 35 B) 128 C) 30 D) 11
3. $3x - 1 < \frac{2024}{3x+1}$ tengsizlikni qanoatlantiradigan eng katta x butun sonni toping.
A) 15 B) 14 C) -16 D) -15
4. ABC uchburchakning BK bissektrisasi o'tkazilgan va $\angle AKB = 2\angle BKC$. ABC uchburchakning C va A burchaklari ayirmasini toping.
A) 30° B) 40° C) 60° D) 80°
5. Ifodani soddalashtiring: $2\sin 40^\circ + 2\cos 130^\circ + \sin 160^\circ - \cos(-110^\circ)$
A) 0 B) $2\sin 70^\circ$ C) $2\sin 40^\circ$ D) $2\sin 20^\circ$
6. $\log_x \frac{19}{8} < \log_x \frac{11}{5}$ tengsizlikni yeching.
A) $(0; 1]$ B) $(0; 1)$ C) $(1; +\infty)$ D) $(0,5; 1)$
7. $f(x) = 2x^4 + (a - 11)x^3 + 1$ funksiya juft funksiya bo'lsa, $f(1)$ ning qiymatini toping. A) 11 B) -8 C) 2 D) 3
8. ABC uchburchakda: $2\sqrt{3}\cos A = 2\cos B = \sqrt{3}$ tenglik o'rinli bo'lsa, $\angle ACB$ ni toping.
A) 30° B) 60° C) 120° D) 90°
9. 9 raqami bilan tugaydigan 9 ga karrali nechta uch xonali son mavjud?
A) 10 B) 9 C) 8 D) 7
10. Ifodani soddalashtiring $\log_{25}\sqrt{5} + \log_6^2 3 \cdot \log_3 6 + \log_6 2$
A) 1 B) $1,25 + \log_6 3$ C) $0,25 + \log_6 12$ D) 1,25

2-qism: Har bir topshiriq 1,5 ball dan baholanadi

11. Tenglamaning haqiqiy ildizlari sonini toping.
 $(2^{x-1} - 1)(|x + 1| - 2)(\log_2 x - 1) = 0$
A) 1 B) 2 C) 3 D) $\emptyset$
12. Burchaklari $15^\circ, 60^\circ, 105^\circ$ bo'lgan uchburchakka tashqi chizilgan aylana radiusi $2\sqrt[4]{3}$ ga teng. Uchburchak yuzini toping.
A) 3 B) $2\sqrt{3}$ C) 2 D) $3\sqrt{3}$
13. x va y sonlari $x(x - y) = y(x + y) = 1$ tenglikni qanoatlantiradi. $xy(x^4 - y^4)$ ifodaning qiymatini toping A) 1 B) 2 C) 4 D) 8
14. $\cos(-2024^\circ) \cdot \sqrt{\cos^{-2} 2024^\circ} = ?$ A) 1 B) -1 C) $-\operatorname{tg} 44^\circ$ D) $-\operatorname{tg} 56^\circ$
15. ABCD kvadratning AD tomonida K nuqta belgilandi, AB nurning B dan keyin davomida L nuqta olindi. Agar $\angle KCB = 60^\circ, KD = 2\sqrt{2}, LB = 1$ ekani ma'lum bo'lsa, LC ni toping. A) 5 B) 4 C) $\sqrt{26}$ D) $\sqrt{33}$



16. Agar $f(1-x) = 1-x^2$ bo'lsa, $f(-1) + f(0) + f(1)$ yig'indini toping.
A) 0 B) -1 C) 1 D) -2
17. $9 \cdot 99 \cdot 999 \cdot \dots \cdot \underbrace{999 \dots 9}_{2024}$ ko'paytmani 1000 ga bo'lgandagi qoldiqni toping.
A) 9 B) 109 C) 891 D) 991
18. Tengsizlikning butun yechimlari sonini toping. $\sqrt{28-x^2} + \sqrt{-x} \geq 4$
A) 4 B) 5 C) 6 D) 11
19. O markazli aylananing OA, OB, OC radiuslari o'tkazilgan. Agar $\overrightarrow{OA} + \overrightarrow{OB} = \overrightarrow{OC}$ bo'lsa, AOB burchakni toping. A) 90° B) 30° C) 120° D) 60°
20. $5^{2024} - 3^{2024}$ ayirmani 34 ga bo'lgandagi qoldiqni toping.
A) 24 B) 2 C) 18 D) 0

3-qism: Har bir topshiriq 2,6 balldan baholanadi

21. Soddashtiring: $\left(\frac{3}{\sqrt[3]{64}-\sqrt[3]{25}} + \frac{\sqrt[3]{40}}{\sqrt[3]{8}+\sqrt[3]{5}} - \frac{10}{\sqrt[3]{25}} \right)^{-1} \cdot (13 - 4\sqrt[3]{5} - 2\sqrt[3]{25}) + \sqrt[3]{25}$
22. Agar $a^2 + a + 11 = 6\sqrt{a^2 + a + 2}$ bo'lsa, $a - \frac{7}{a}$ ifodaning qiymatini toping.
23. Agar $\operatorname{tg} \alpha + \sin \alpha = 1$ ($0 < \alpha < \frac{\pi}{2}$) bo'lsa, $(\sin 2\alpha + 2)^2$ ni hisoblang
24. O'nli yozuvida hech bo'lmasa bitta juft raqam bo'lgan to'rt xonali sonlar nechta?
25. Radiusi 16 ga teng bo'lgan aylanaga ichki chizilgan qavariq to'rtburchakning uchta tomoni 8 ga teng bo'lsa, uning to'rtinchi tomonini toping.
26. a, b, c — turli raqamlar, $\overline{aabc}$ — to'rt xonali son va $\overline{aabc} = (a + b + c)^3$ bo'lsa, $a + b - c$ ni hisoblang.
27. Balandligi $\sqrt{19} + 2$ ga, yoyi 120° ga teng bo'lgan segmentga kvadrat ichki chizilgan. Kvadrat yuzini toping.
28. $\frac{x^2-3px+2p^2}{x+2p-6} = 0$ tenglama faqat bitta ildizga ega bo'ladigan p parametrning nechta qiymati mavjud?
29. $20 \cdot [u] = 24 \cdot \{u\}$ tenglamaning ildizlari yig'indisini toping, bunda $[u]$ — u sonining butun qismi, $\{u\}$ — u sonining kasr qismi.
30. Dskada 36 soni yozilgan. Har minutda doskadagi son o'chiriladi va uning o'rniga o'chirilgan son raqamlari ko'paytmasiga 13 ni qo'shib, hosil qilingan son yoziladi. bir soatdan keyin doskada qaysi son hosil bo'ladi?



Yopiq turdagi (1–20) test topshiriqlari.

1. [0,9 ball] a , b haqiqiy sonlar uchun

$\frac{a+2b}{a-2b} = 3$ bo'lsa, $\frac{a+3b}{a-3b}$ ning qiymatini toping.

A) 5 B) 6 C) 7 D) 8

2. [0,9 ball] Ichida 4 ta kitob bo'lgan qutining vazni 10 kg va ichida 6 ta kitob bo'lgan xuddi shu qutining vazni 13 kg. Agar barcha kitoblarning vazni bir xil bo'lsa, bo'sh qutining vaznini toping.

A) 1 B) 2 C) 3 D) 4

3. [0,9 ball] 9800 sonining nechta to'la kvadrat bo'luvchisi bor?

A) 4 B) 8 C) 12 D) 36

4. [0,9 ball] Bir bazmda, har bir bola bittadan olma olsa, savatda yana 7 ta olma ortib qoladi. Agar har bola ikkitadan olma olsa, 16 ta olma yetmay qoladi. Bazmda nechta bola bo'lgan?

A) 23 B) 16 C) 14 D) 7

5. [0,9 ball] $x < -1$ haqiqiy son uchun quyidagi javoblardan qaysi biri to'g'ri?

A) $x^3 < x < x^2$ B) $x < x^2 < x^3$
C) $x < x^3 < x^2$ D) $x^2 < x^3 < x$

6. [0,9 ball] $a_1, a_2, a_3, \dots, a_n$ arifmetik progressiya uchun $a_5 + a_8 + a_{11} = 12$ va $a_7 + a_{10} + a_{13} = 18$ shartlar o'rinli. Agar $a_k = 5$ bo'lsa, k ning qiymatini toping.

A) 12 B) 11 C) 10 D) 9

7. [0,9 ball] Bugun haftaning dushanba kuni va Asadbek bugun 290 betli kitobni o'qishni boshlaydi. U haftaning dushanba kunlari 25 bet va haftaning boshqa kunlarida 4 betdan kitob o'qisa, haftaning qaysi kunida kitobni o'qib tugatadi?

A) Payshanba B) Juma
C) Shanba D) Yakshanba

8. [0,9 ball] 2000 va 3000 orasidagi natural sonlarning nechtasining raqamlari yig'indisi 5 ga teng.

A) 6 B) 10 C) 15 D) 21

9. [0,9 ball] Bir futbol o'yinidagi yakuniy hisob 3:2 bilan yakunlandi. O'yinning birinchi yarmidagi yakuniy hisobning bo'lishi mumkin bo'lgan qiymatlari sonini toping.

10. [0,9 ball] Imtixon 20 ta savoldan iborat.

Har bir to'g'ri javob uchun 3 ball beriladi, har noto'g'ri javob uchun 1 ball ayiriladi va belgilanmagan javob uchun hech qanday ball ayirilmaydi va qo'shilmaydi. Asad 15 ta savolga javob berdi va qolganlarini bo'sh qoldirdi. Agar u to'plagan ball 17 dan past bo'lsa, u eng ko'pi bilan nechta savolga to'g'ri javob bergan?

A) 7 B) 8 C) 9 D) 10

11. [1,5 ball] f funksiya uchun

$f(x) = 1 - f(x - 1)$ va $f(4) = 6$ bo'lsa, $f(10)$ ning qiymatini toping.

A) 6 B) -5 C) 12 D) 8

12. [1,5 ball] $1 - \sin x + \sqrt{3y - x} = 0$

tenglama orqali $\frac{6(x-y)}{\pi}$ ning qiymatini toping, bunda $0 \leq x \leq \pi$.

A) 4 B) 3 C) 2 D) 1

13. [1,5 ball] x, y – musbat haqiqiy sonlar

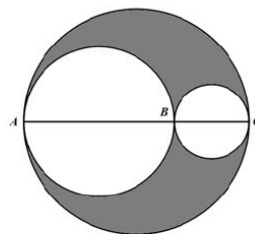
uchun $\frac{1}{8} < \frac{\sqrt{x}}{x} \leq \frac{1}{3}$ va $\frac{2}{5} < 1 - \frac{\sqrt{y}}{y} \leq \frac{1}{2}$

tengsizliklar o'rinli bo'lsa, $x - y$ ifodaning qabul qilishi mumkin bo'lgan eng kichik qiymatini toping.

A) 4 B) 5 C) 6 D) 8

14. [1,5 ball] AB , AC va BC lar berilgan chizmadagi uchta aylanalar diametrlari ekanligi ma'lum va $AB:BC = 3:2$.

Bo'yalmagan qism katta aylananing necha foizini tashkil qiladi?



A) 64 % B) 60 % C) 56 % D) 52 %

15. [1,5 ball] a , b va c turli natural sonlar

uchun $\frac{6^{a \cdot 15^b}}{9^{b \cdot 10^c}}$ soni natural son bo'lsa, sonlarni o'sish tartibida iovlashtiring.



Agar quyidagilardan bittasi toping.

A) 12 B) 7 C) 6 D) 5

Matematika 11

A) $c < b < a$ B) $c < a < b$
C) $b < c < a$ D) $a < b < c$

~ 2 ~

Matematika

16. [1,5 ball] Agar tub sonning birlar xonasidagi raqamini olib tashlashimizdan hosil bo'lgan son ham tub son bo'lsa, biz bu sonni *tub – tub* son deymiz. Masalan 317 - *tub – tub* son, chunki 317 ham 31 ham tub sonlar. Nechta ikki xonali *tub – tub* sonlar mavjud?

A) 7 B) 8 C) 9 D) 10

17. [1,5 ball] $x^2 + ax + b = 0$ va $x^2 + ax + b + 1 = 0$ kvadrat tenglamalar ikkalasi ham butun ildizlarga ega bo'ladigan va $0 \leq b \leq 10$ shartni qanoatlantiradigan barcha $(a; b)$ butun juftliklar sonini toping.

A) 3 B) 4 C) 6 D) 7

18. [1,5 ball] Anvarning sumkasida 10 ta ko'k, 6 ta yashil va 5 ta qizil rangli sharlar bor. U har safar sumkasidan bitta shar oladi va stol ustiga qo'yadi. Stolda har bir rangli shardan kamida

bittadan bo'lishini kafolatlash uchun u sumkasidan eng kamida nechta shar olishi kerak?

A) 12 B) 15 C) 16 D) 17

19. [1,5 ball] $ABCD$ to'g'ri to'rtburchakda $\angle BAD$ ning bissektisasi BC tomonni P nuqtada kesadi. Agar $AP:PD = 4:3$ bo'lsa, $PD:PC$ nisbatni toping.

A) 2:1 B) 2,5:1 C) 3:1 D) 3,5:1

20. [1,5 ball] Agar arifmetik progressiyaning uchinchi va to'rtinchi hadlari mos ravishda 3 va 8 ga oshirilsa, progressiyaning birinchi to'rtta hadi geometrik progressiyaning tashkil qiladi. Arifmetik progressiyaning uchinchi hadini toping.

A) 12 B) 9 C) 5,5 D) 1,5

Ochiq turdagi (21–30) test topshiriqlari.

21. [2,6 ball] $1^{2025} + 2^{2025} + 3^{2025} + \dots + 2025^{2025}$ ifodaning oxirgi raqamini toping.

Javob: _____

22. [2,6 ball] x – haqiqiy son uchun $(4 - x)(2 - x)(x + 4)(x + 2)$ ifodaning eng kichik qiymatini toping.

Javob: _____

23. [2,6 ball] $ABCD$ kvadrat tashqarisida E nuqta olingan, bunda $\angle DCE = 51^\circ$ va $\angle CDE = 78^\circ$. $\angle DAE$ ni toping.

Javob: _____



24. [2,6 ball] 20 nafar o'quvchi futbol, basketbol yoki badminton sport turlarini tanlashi mumkin. Har bir o'quvchi kamida ikkita sport turini tanlashi kerak. 14 nafar o'quvchi futbol, 15 nafar o'quvchi



On o'quvchi kamida ikkita sport turini tanlashi kerak. 14 nafar o'quvchi futbol, 15 nafar o'quvchi basketbol va 16 nafar o'quvchi badminton o'ynashni tanlagan bo'lsa, uchchala sport turini tanlagan o'quvchilar sonini toping.

Javob: _____

~ 3 ~

Matematika

25. [2,6 ball] $ABCD$ to'rtburchakda $AB = CD$, $\angle ADC = 63^\circ$, $\angle DCA = 41^\circ$ va $\angle ACB = 104^\circ$ bo'lsa, $\angle BAC$ ni toping.

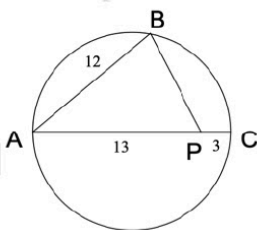
Javob: _____

26. [2,6 ball]

$\left(2 - \frac{1}{2}\right)\left(3 - \frac{1}{3}\right)\left(4 - \frac{1}{4}\right) \dots \left(60 - \frac{1}{60}\right) = \frac{a!}{b!}$ tenglik orqali $a + b$ ning qiymatini toping.

Javob: _____

27. [2,6 ball] Chizmada uzunligi 16 cm bo'lgan AC diametrda P nuqta olingan, bunda $PC = 3$ cm. Agar aylanadagi B nuqta uchun $AB = 12$ cm bo'lsa, BP kesmaning uzunligini (cm) toping.



Javob: _____

28. [2,6 ball] x_n ketma-ketlik uchun $x_1 = 20$, $x_2 = 101$ va ixtiyoriy $n > 2$ uchun $x_n = \frac{x_{n-1} + 1}{x_{n-2}}$ shartlar o'rinli. x_{2025} ning qiymatini toping.

Javob: _____

29. [2,6 ball] Tenglamaning barcha ildizlari yig'indisini toping: $[3x + 1] = 2x - \frac{1}{2}$, bunda $[x]$ — x ning butun qismi.



Javob: _____